

Artificial Intelligence

Lesson 18: Probability Foundations

Represent → Combine → Condition → Compare

Uncertain system state

Good lighting .60	Poor lighting .40
Confident .56	Uncertain .44

Probability is a language for reasoning when the state is uncertain.



From known states to uncertain states

When the agent is uncertain, it stores belief across possibilities.



Instead of storing one state, an uncertain agent may store a probability distribution.

Today's mission

Probability operations answer different questions about uncertainty.

1

Interpret random variables and probability distributions.

2

Compute joint, marginal, and conditional probabilities.

3

Apply product rule and marginalization.

4

Determine independence and recognize conditional independence.

Working rule: Compute → Interpret → Compare

Random variable vs. value

Use notation to separate the uncertain quantity from one possible assignment.

L = lighting quality

Random variable

The aspect of the world we are uncertain about.

L = Good

L = Poor

Values / assignments

Specific possible outcomes for that variable.

A probability model assigns numbers to assignments.

Distribution vs. single probability

A distribution is a whole table; a probability is one entry.

$P(L)$

L	$P(L)$
Good	.60
Poor	.40

Distribution

All probabilities for all values of L .

$$P(L = \text{Good}) = .60$$

Single probability

One number from the distribution.

This distinction matters later when $P(X_t)$ means an entire belief distribution over time.

Probability describes uncertainty—not certainty

A probability is not a guarantee about the next single outcome.

Interpret:

$$P(L = \text{Good}) = .60$$

Good interpretation

Given the model and available information, we assign 60% probability to good lighting across comparable cases.

Common mistake

“The lighting is 60% good.”

No, the actual lighting condition is one value; our belief is uncertain.

A probability summarizes uncertainty before the outcome is known.

You Try: valid distribution?

Check whether the table could represent a probability distribution.

State	P(State)
Normal	.70
Degraded	.20
Failed	.15

A

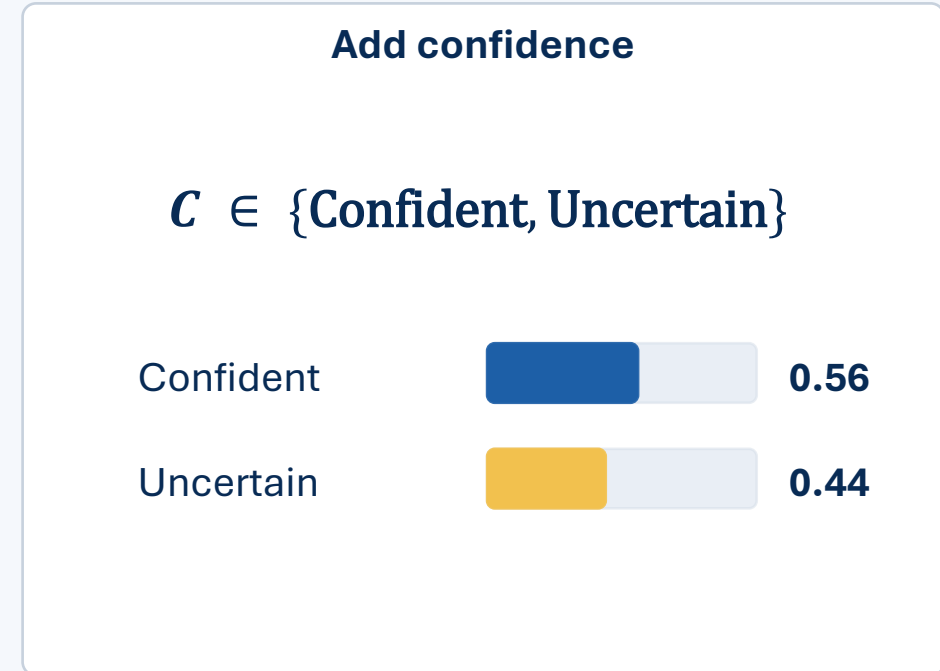
State	P(State)
Normal	.70
Degraded	.20
Failed	.10

B

Which table is valid? Explain using the two distribution rules.

One variable is often not enough

Many AI judgments depend on combinations of uncertain variables.



Next question: what is the probability of lighting and confidence together?

The joint distribution describes combinations

$P(L, C)$ assigns probability to each complete assignment.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

$$P(L, C)$$

Assume L and C are dependent – this table is simply given (as with quizzes & GR)

Every row is one possible complete combination.

Example assignment:

$L = \text{Good}$
 $C = \text{Confident}$

Joint probability asks how likely a combination is.

You Try: read the joint table

No formula yet—just read the model.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

Answer from the table:

1. $P(L = \text{Good}, C = \text{Confident}) = \underline{\hspace{2cm}}$

2. $P(L = \text{Poor}, C = \text{Uncertain}) = \underline{\hspace{2cm}}$

3. Interpret one answer in words.

Joint = both conditions hold.

Events are questions about outcomes

An event can contain one outcome or many outcomes.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

Good AND Confident

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

Good lighting

Many useful probability questions are about sets of rows in a table.

Marginalization: ignore what you do not care about

Keep the variable you care about; sum over the possibilities you do not.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32



$$P(C = \text{Confident})$$

$$.48 + .08 = .56$$

We summed over both lighting values because the question did not ask about lighting.

Marginalization formula

The formula is a shorthand for summing matching rows.

Marginalization

$$P(C) = \sum_l P(L, C)$$

Sum over every value of L to get a distribution over C .

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

$$P(C = \text{Confident}) = P(\text{Good}, \text{Confident}) + P(\text{Poor}, \text{Confident})$$

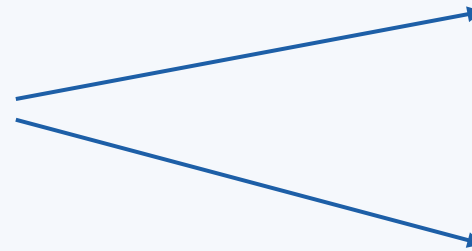
$$= .48 + .08 = .56$$

Visual marginalization

The joint table can collapse into smaller distributions.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

Joint model



Marginal distributions

L	$P(L)$
Good	.60
Poor	.40

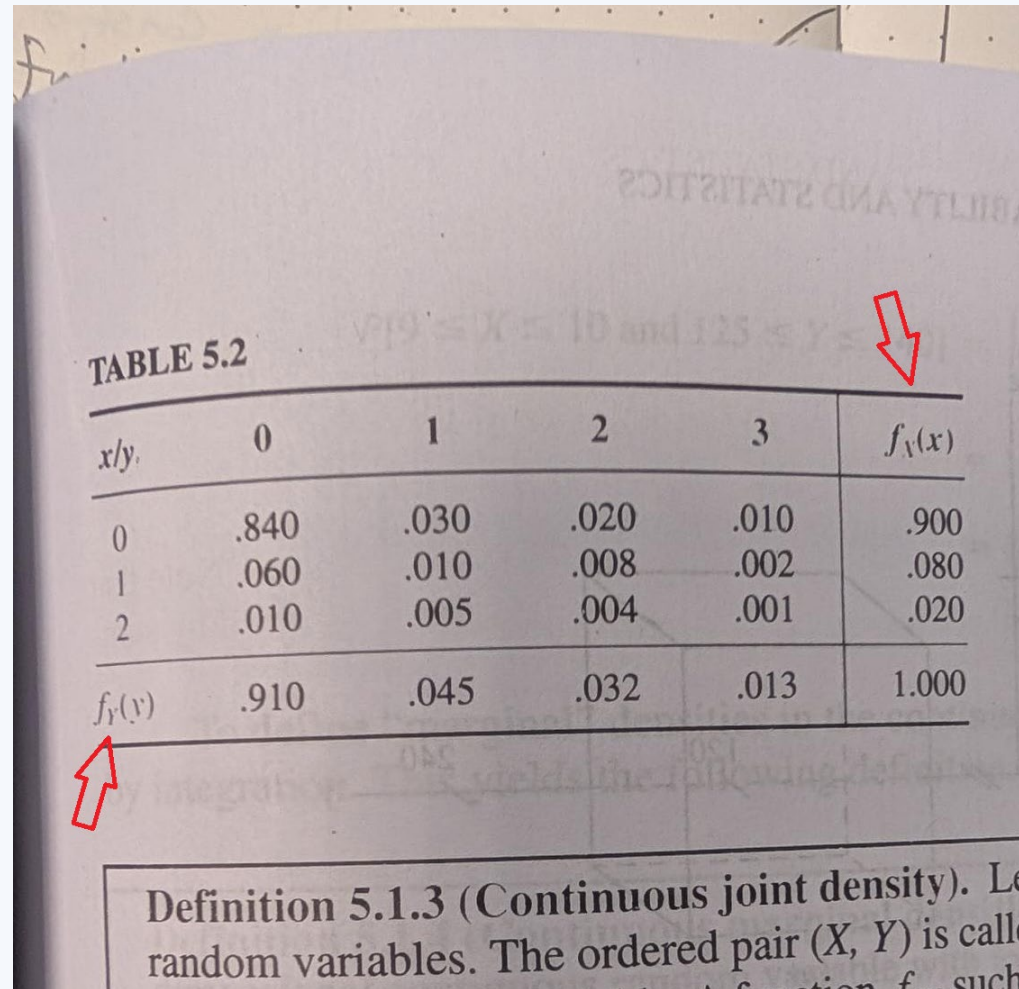
C	$P(C)$
Confident	.56
Uncertain	.44

Marginalization removes a variable by adding over its possible values.

Visual marginalization

Often found in the MARGINS of a table – hence the name!

TABLE 5.2



The image shows a photograph of a textbook page. A joint probability table is presented with two red arrows pointing to the marginal distributions. One arrow points to the column header $f_X(x)$ and the other points to the row header $f_Y(y)$. The table is titled 'TABLE 5.2' and contains numerical values for a joint distribution of two discrete random variables, X and Y.

x/y	0	1	2	3	$f_X(x)$
0	.840	.030	.020	.010	.900
1	.060	.010	.008	.002	.080
2	.010	.005	.004	.001	.020
$f_Y(y)$	.910	.045	.032	.013	1.000

Definition 5.1.3 (Continuous joint density). Let X and Y be continuous random variables. The ordered pair (X, Y) is called a continuous random vector if there exists a nonnegative function $f_{X,Y}$ such that

You Try: sum out a variable

Use a different table; answer the questions.

	Confident	Uncertain
Good	.18	.29
Poor	.16	.37

Pairs: compute and interpret

1. $P(C = \text{Uncertain}) = \underline{.29 + .37 = .66}$

2. $P(L = \text{Poor}) = \underline{.16 + .37 = .53}$

3. What variable did you “sum out” in each case?

$$P(C) = \sum_L P(L, C)$$

Evidence changes the population

Conditioning restricts the world before we calculate.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

Suppose we learn:

$$L = \text{Good}$$

Rows where $L = \text{Poor}$ no longer match the evidence.

The remaining probability mass is $.48 + .12 = .60$, not 1.

Conditional probability: restrict, then renormalize

A conditional probability asks about a smaller evidence-consistent world.

1. Restrict

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

2. Renormalize

$C \mid L = \text{Good}$	P
Confident	$.48 / .60 = .80$
Uncertain	$.12 / .60 = .20$

$$P(C = \text{Confident} \mid L = \text{Good}) = .80$$

Conditioning turns surviving probability mass into a new distribution that sums to 1.

Conditional probability formula

Now the formula names what we just did visually.

Conditional probability

$$P(A \mid B) = P(A, B) / P(B)$$

Of the probability mass where B is true,
what fraction also has A ?

Running example

$$\begin{aligned} P(C = \text{Confident} \mid L = \text{Good}) \\ &= \frac{P(\text{Good, Confident})}{P(\text{Good})} \\ &= \frac{.48}{.60} = .80 \end{aligned}$$

Read “ $\mid$ ” as “given.” It marks the evidence we are conditioning on.

You Try: restrict + renormalize

Compute a conditional probability from the same model.

	Confident	Uncertain
Good	.18	.29
Poor	.16	.37

$$.18 + .16 = .34$$

Pairs: compute and interpret

Compute:

$P(L = \text{Good} \mid C = \text{Confident})$

1. Which column survives?
2. What is the denominator?
3. What does the answer mean?

$$.18 * .34 \approx .53$$

Hint: answer should not be the joint probability .18.

Joint versus conditional

These look similar but answer different questions.

Joint

$$P(L=\text{Good}, C=\text{Confident}) = .48$$

Out of all possible cases, how often are both true?

Conditional

$$P(C=\text{Confident} \mid L=\text{Good}) = .80$$

Among cases with good lighting, how often is confidence high?

Same row can support different questions depending on the denominator.

Product rule: reconstruct the joint

The product rule reverses the conditional-probability relationship.

Product rule

$$P(A, B) = P(B)P(A | B)$$

Use a marginal and a conditional to reconstruct a joint probability.

Running example

$$P(L=\text{Good}) = .60$$

$$P(C=\text{Confident} | L=\text{Good}) = .80$$

$$P(\text{Good, Confident}) = (.60)(.80) = .48$$

The product rule tells you how the joint table could have been built.

General rule versus independence shortcut

Multiplying marginals is only valid when independence is justified.

$$P(A, B) = P(A)P(B | A)$$

Always valid

Uses a conditional probability.

May also see it written this way

$$P(A \cap B) = P(A)P(B)$$

$$P(A, B) = P(A)P(B)$$

Shortcut

Only if A and B are independent.

Do not assume independence because multiplication looks easier.

Independence asks whether evidence changes belief

If observing B changes belief about A, the variables are dependent.

$$P(C = \text{Confident}) = .56$$

Before evidence

$$P(C = \text{Confident} \mid L = \text{Good}) = .80$$

After learning $L = \text{Good}$

$$.80 \neq .56$$

Learning about lighting changes our belief about confidence, so L and C are dependent.

Tests for independence

Two equivalent checks are useful in different situations.

Conditional test

$$P(A \mid B) = P(A)$$

Evidence B does not change belief in A.

Product test

$$P(A, B) = P(A)P(B)$$

The joint equals the product of marginals.

Both tests express the same idea: one variable gives no information about the other.

You Try: does evidence matter?

Compare marginal and conditional probabilities.

	Confident	Uncertain
Good	.48	.12
Poor	.08	.32

Pairs: decide independence

Use: $P(A | B) = P(A)$

$P(C = \text{Confident}) \approx .56$

$P(C = \text{Confident} | L = \text{Good}) \approx .29$

Are L and C independent?

Justify in one sentence.

Answer format: They are independent/dependent because _____.

Independence does not mean...

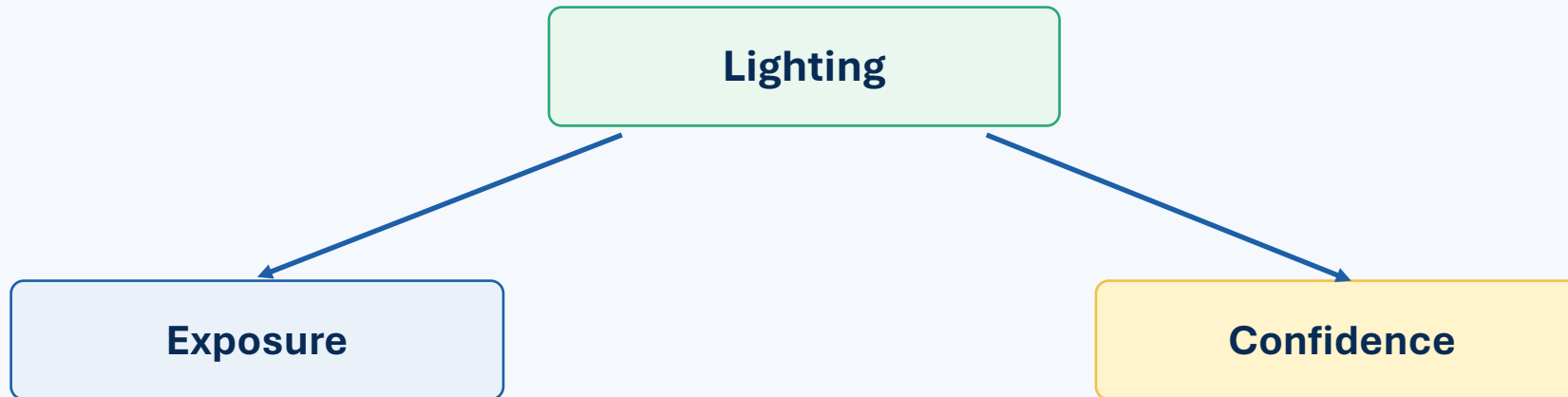
Misconceptions usually come from treating independence as “separate events.”

Claim	Independence?
Events cannot both occur	False
The probabilities are equal	False
One event has probability zero	False
Learning one does not change belief in the other	True

Memory cue: independence is about information, not physical separation.

Conditional independence preview

Sometimes dependence disappears after another variable is known.



Exposure and confidence may appear related because both depend on lighting.

But once lighting is known, exposure may add no new information about confidence.

$$E \perp C \mid L$$

We will use graphs to reason about this in L20.

Integrated practice: one table, four questions

Use one representation to answer multiple probability questions.

	Confident	Uncertain
Good	.20	.30
Poor	.05	.45

Answer as pairs

1. Joint: $P(L = \text{Poor}, C = \text{Uncertain})$
2. Marginal: $P(C = \text{Uncertain})$
3. Conditional: $P(C = \text{Confident} \mid L = \text{Good})$
4. Independence: Are L and C independent? Why?

For each answer: compute the number and interpret the question it answers.

Compute → Interpret → Compare

Probability answers are incomplete until students can explain what they mean.

Compute

What is the numerical result?

Interpret

What event or question does the number describe?

Compare

How is it different from a related probability?

Example: $P(C = \text{Confident} \mid L = \text{Good}) = .80$

Interpretation: among cases with good lighting, the model assigns 80% probability to a confident vision output.

Compare: this is different from $P(C = \text{Confident}) = .56$ because evidence changed the population.

Transfer problem: robot localization

Same tools, new context.

A delivery robot records terrain quality T and localization quality Q .

Terrain T	Localization Q	$P(T,Q)$
Smooth	Reliable	.45
Smooth	Unreliable	.15
Rough	Reliable	.10
Rough	Unreliable	.30

Compute

1. $P(Q = \text{Reliable})$
2. $P(Q = \text{Reliable} \mid T = \text{Smooth})$
3. Does smooth terrain change belief about reliable localization?

Transfer check: use the same operations without the original story.

Diagnose the mistake

Correcting errors is often the fastest way to clarify the concept.

Claim	Correction
$P(A,B)=P(A)+P(B)$	
$P(A B)=P(B A)$	
Marginalization multiplies across table cells	
$P(A)=P(B)$ proves independence	
$P(A B)=P(A)$ is evidence of independence	

Ask: What question is this probability notation actually answering?

Probability toolbox

Each tool answers a different kind of question.

Question	Tool
What combinations are possible?	Joint distribution
I do not care about one variable.	Marginalize / sum out
I observed evidence.	Condition / renormalize
I know a conditional and marginal.	Product rule
Does evidence change belief?	Independence test

Use the question to choose the operation—not the other way around.

Exit ticket: bridge to Bayes' Rule

Complete the sentence, then preview the next problem.

Marginalization means _____.

Conditioning means _____.

A and B are independent if _____.

Today we computed $P(C | L)$.



Next: what if we need $P(L | C)$?

Bayes' Rule lets us reverse the direction of conditioning and update belief from evidence.